

Problem Statement 1 – Unit Testing FastAPI User Authentication API

Scenario

You are developing a FastAPI backend for a simple **User Authentication System**. The API allows users to **register**, **login**, and access a **protected profile endpoint**.

API Endpoints

- POST /register – Register a new user
- POST /login – Authenticate user and return JWT token
- GET /profile – Return logged-in user details (protected)

Requirements

- Passwords must be hashed
- Login should return a JWT token
- /profile should be accessible only with a valid token

Testing Task

Write **unit tests using pytest** to verify:

1. User registration succeeds with valid input
2. Duplicate email registration fails
3. Login succeeds with correct credentials
4. Login fails with invalid password
5. Protected endpoint rejects requests without token
6. Protected endpoint works with valid token

Goal of Testing

- Validate API logic
- Verify authentication flow
- Ensure security rules are enforced

REACT FRONTEND – PROBLEM STATEMENTS (Testing the SAME FastAPI)

Problem Statement 2 – Unit Testing React Product List Screen

Scenario

A React application displays a list of products fetched from the FastAPI backend using Axios.

UI Features

- Fetch products from GET /products
- Display product name, price, and category
- Show loading indicator while fetching
- Show error message on API failure

Testing Task (Using Jest + React Testing Library)

Write unit tests to verify:

1. API call is triggered on component load
2. Loading message is shown while data is fetched
3. Product list is rendered after successful API response
4. Error message is displayed when API fails
5. Empty product list message is shown when no data exists

Goal of Testing

- Ensure correct API integration
- Validate UI behavior for different API states
- Prevent UI regressions